

Introduction chemical modelling with **ORCHESTRA**

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Introduction chemical modelling with ORCHESTRA

Hans Meeussen NRG/TU Delft, 18/09/2024

Part 1: beginners

- ORCHESTRA, what is different (compared to PHREEQC, ECOSAT, GEMS, MINTEQA6, REAKTORO, GWB ..)
- Download “install” ORCHESTRA
- Define a chemical equilibrium system input, output
- Make predominance graphs
- Questions

Part 2: more experienced users

- A bit more on structure
- How to work with “phases”, “entities”, “reactions”
- How to work with adsorption models
- Questions

Introduce myself

Hans Meeussen

- Research scientist at NRG (Nuclear Research and consultancy Group) Petten
- Part time associate professor TU Delft, department of Civil Engineering

Education

- MSc, PhD Wageningen, Soil chemistry / Software engineering

Research interests

- Modelling reactive transport processes
- Radionuclide migration in rock, soil and concrete
- Model development

Involvement in chemical model development started in Wageningen University ca. 1990 (ECOSAT).
(group Willem van Riemsdijk, developing new adsorption models CD-MUSIC, NICA Donnan, LCD etc.)

ORCHESTRA development started ca. 1994, Macaulay (now Hutton) institute Aberdeen, UK.

ORCHESTRA, what is similar to other codes...

- Can solve chemical equilibrium systems
- Same application area as PHREEQC, ECOSAT, GEMS, MINTEQA2, REAKTORO, GWB ..
- Can read PHREEQC thermodynamic database files
- Define chemical system interactively

ORCHESTRA, what is different to other codes...

ORCHESTRA, what is similar to other codes...

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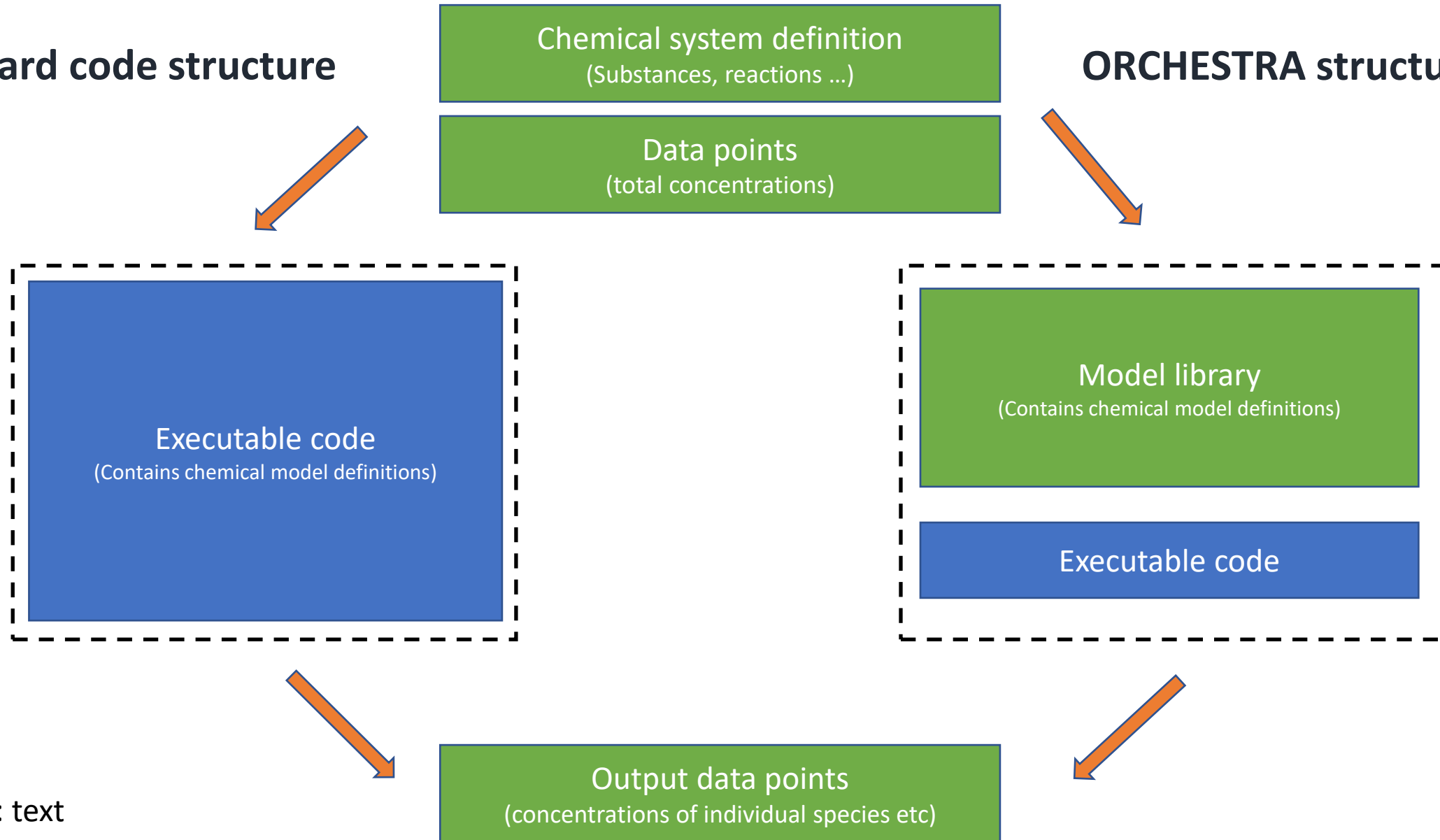
ORCHESTRA, what is different to other codes...

→ Internal structure

Comparison of standard code structure with ORCHESTRA structure

Standard code structure

ORCHESTRA structure



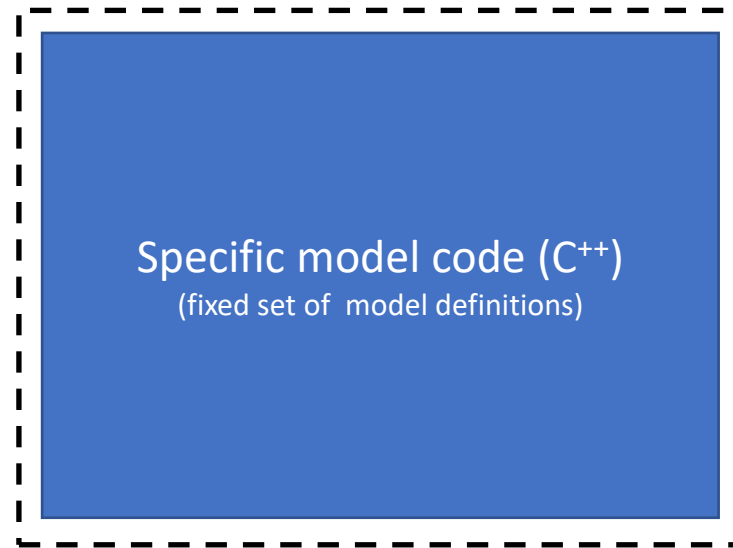
 : text
 : executable code

Comparison of standard solver structure with ORCHESTRA solver structure

Standard code structure

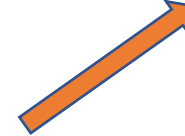
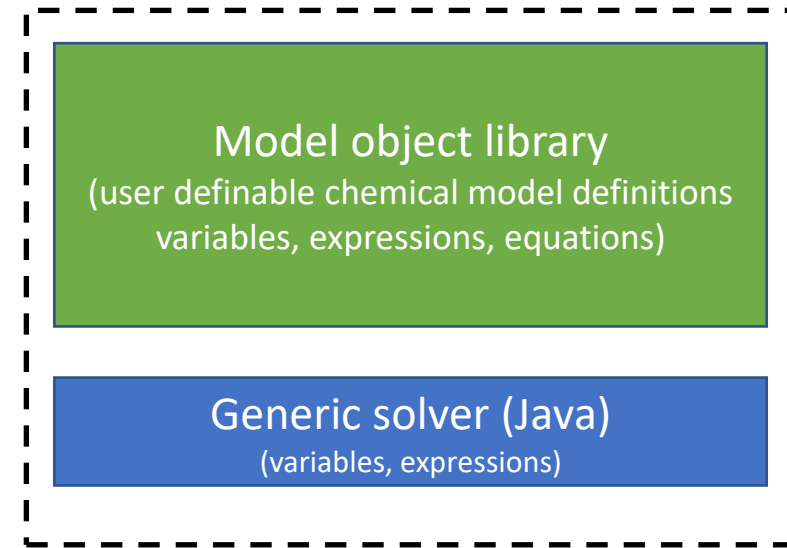
(e.g. PHREEQC, GEMS, GWB, Visual Minteq, HYTEC, ECOSAT)

Chemical model definitions **inside** executable



ORCHESTRA code structure

Chemical model definitions in **external text** file(s)



Same functionality

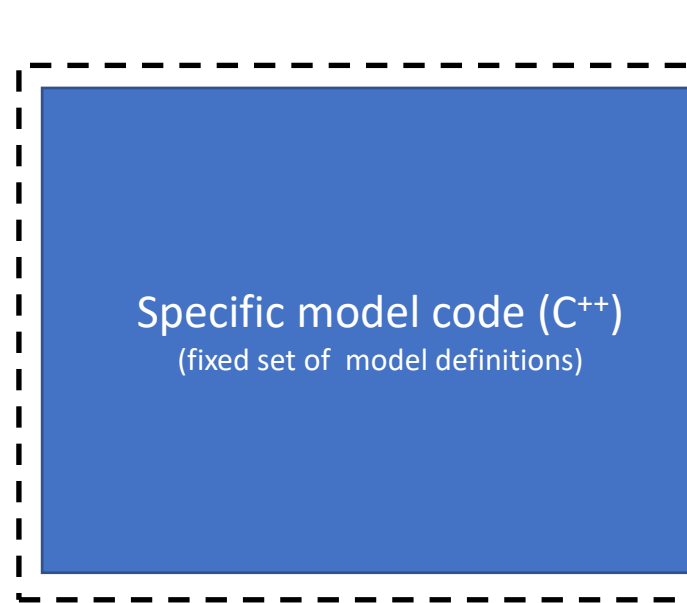
 : text
 : source code

Comparison of standard solver structure with ORCHESTRA solver structure

Standard code structure

(e.g. PHREEQC, GEMS, GWB, Visual Minteq, HYTEC, ECOSAT)

Chemical model definitions **inside** executable

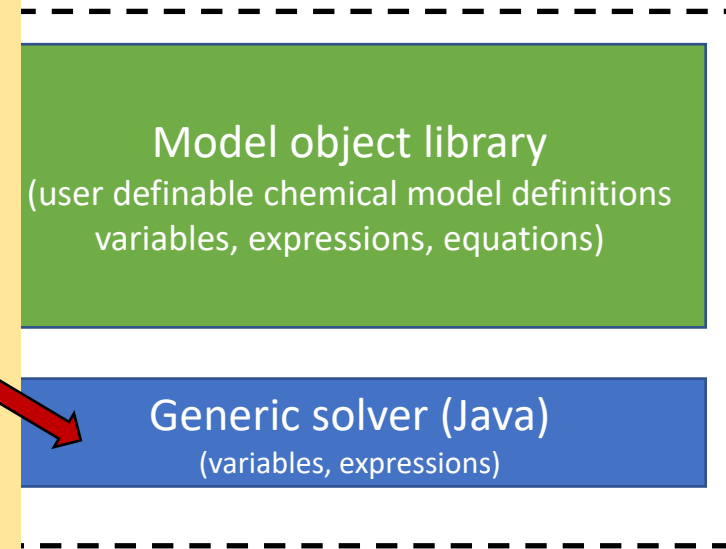


ORCHESTRA solver
can use data
(numbers) but **also**
code (variables,
expressions,
equations) as input!

Databases can
contain code as well
as parameters

ORCHESTRA code structure

Chemical model definitions in **external text file(s)**



Same functionality

 : text
 : source code

ORCHESTRA: all chemical “knowledge” present in input text file(s)

Model object library text file contains definitions of e.g.:

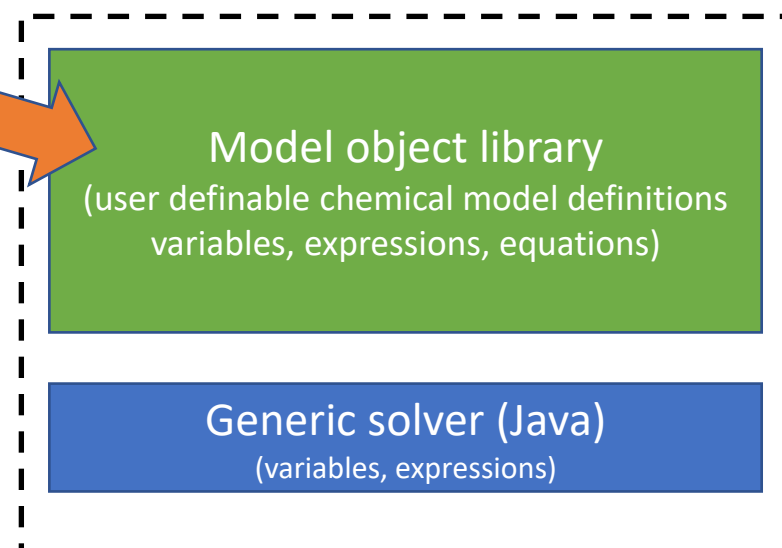
- Reaction
- Master species
- Aqueous species
- Activity correction models (Davies, Debye Hueckel, SIT etc.)
- Solid Phase
- Gas Phase
- Solid solutions
- Ion exchange Vanselow Gaines Thomas, etc..
- Surface complexation models (Basic stern, Dzombak and Morel, CD Music, Bradbury Bayens.....)
- NICA – Donnan distributed log K model
-

All ORCHESTRA “chemical knowledge” is concentrated here.

www.meeussen.nl/orchestra/objects2024.txt

Advantages:

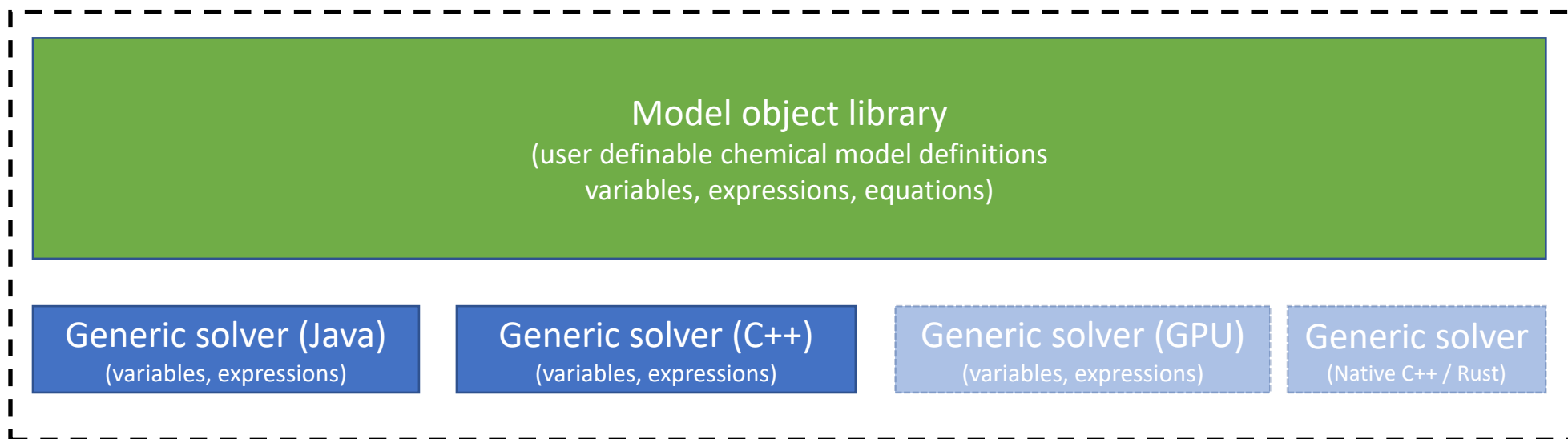
- **Users** can **change**/add models **without** the need to **change** the solver **source** code
- Solver **executable** is very **small** and **simple**



 : text

 : source code

The structure of ORCHESTRA allows to update the solver independently of the chemical model library (and vice versa)



ORCHESTRA solver structure allows:

- To **change chemical models without changing executable** (useful for chemical model development)
- To **change executable without changing chemical models** (useful for developing C⁺⁺ version, but also useful for possible future developments)

ORCHESTRA structure advantages

- **Adsorption (and other) models** can be **added by users** without changing source code
- **Modular**: parts can be independently exchanged for newer versions (e.g. **C++** and **Java** version of solver available)
- Solver code is very **flexible, small, and fast** (important for coupling with transport codes and **parallel processing**)
- **Coupling** with external codes (transport/python) is **simple**
- All chemical models are **automatically consistent** with transport coupling
- Well-suited for developing models that are partly “standard geochemistry” combined with specific physical/biological/kinetic models. (especially in case of feed-back between physical and chemical part)

ORCHESTRA structure disadvantages.....

- All chemical models were **implemented newly from scratch**
- ORCHESTRA “language” is **simple**, but **not standard** Java or C++
- Adding models requires **understanding** of “ORCHESTRA language”
- **Flexibility** gives users **freedom** to create their own models (doing this in a **structured way is user responsibility!**)

What does this mean for you?

- **Standard** chemical systems and adsorption models can be defined **interactively** via GUI (Graphical User Interface)
- **Large** library with adsorption models present
- Make use of latest developments
- Freedom to expand existing models with own extensions

However:

- Not all flexibility is accessible via GUI (Graphical User Interface)
- We can switch between GUI and text mode to define or edit parts of the model in text mode
- Text mode offers full flexibility 😊, but more knowledge required 😞
- We focus here on definitions via GUI! 😊

Are there any questions/ comments about structure?

Before we go the demonstration

Installation

1) Install a Java runtime environment. Available for free from: **openjdk.org** or **www.graalvm.org**

2) Download an example folder (that contains all necessary model and input files) from :

www.meeussen.nl/orchestra/getstarted.zip:

www.meeussen.nl/orchestra/getstarted2.zip

Unzip this folder somewhere at your computer

3) Open the folder in the windows file manager and click on **runorchestra.bat** to start the orchestra editor.

or (on Linux, OSX)

Open folder in console and give command: **java -cp orchestra2024.jar orchestra2.composer**

Time for a demonstration!

www.meeussen.nl/orchestra/getstarted.zip

ORCHESTRA

An Object Oriented Framework for Composing
Chemical Speciation and Mass Transport Models

Version: 21 August 2024 07:58

Hans Meeussen
www.meeussen.nl/orchestra
orchestra@meeussen.nl

Intro
Documentation
Chemistry
Input
Output
Output2
Output3
Concert

The ORCHESTRA composer

- Different tabs are connected to set of input (text) files
- These are present in folder (can be edited separately)
- **Composer.inp** defines the files visible in composer
- **Concert.xml** defines calculation structure (input, output files chemical definition file)
- The **chemistry.inp** file contains chemical definition and is edited with GUI
- Input.dat contains input data points
- Output.dat (different versions) contains calculated output in different formats
- Clicking **run** saves all files, runs the concert.xml file and returns after the calculation is ready
- Pocket calculator (uses same calculator as chemistry solver)
- Refresh refreshes databases

ORCHESTRA: Defining a chemical system with the GUI

-creates/edits a text file (chemistry.inp) that contains the complete definition of the chemical system

ORCHESTRA-Composer (Running on Windows 10 with 22 Oracle Corporation, using 16 processing cores)

File Run Tools Help

chemistry1.inp Read Write GUI View

Adsorption models Activity correction Settings Predominance Diagram Output selector

Primary entities/ Master Species Phases & Reactions Variables

Selectable Primary Entities/ Master Species in Database

tot diss min gas

S	M	G	A	
T	N	H	B	Acetate
U	O	I	C	Adipate
Z	P	K	E	Ag
	R	L	F	Al
				Am

Include	Database
☒	thermochimie11a.dat

Selected Primary Entities/ Master Species

Incl.	Primary entity	Phase	Input variable	Fix log activity	Log activity	Concentration	Phase	Expression
☒	CO2[g]	gas		☒	-3.39			
☒	Ca	tot		☐		0.1	tot	
☒	Cl	tot		☐		0.19	tot	
☒	E	tot	pe	☒	-7.5			E.logact = -pe
☒	H	tot	pH	☒	-8.0			H.logact = -pH
☒	O	tot	H2O.logact	☒	0.0			O.logact = H2O.logact

☐ Balance Charge pe

No input file or directory selected.

ORCHESTRA: Defining a chemical system with the GUI

- creates/edits a text file (chemistry.inp) that contains the complete definition of a chemical system
- can be processed by the solver/calculator
- database is only used for selecting substances/reactions in the GUI
- not used during calculations
- input text file can be edited in text mode: click GUI view/text View
- input file is similar to a simple computer program

chemistry1.inp

Read Write GUI View

AdSORPTION models Activity correction Settings Predominance Diagram Output selector

Primary entities/ Master Species Phases & Reactions Variables

Selectable Primary Entities/ Master Species in Database

tot diss min gas

S	M	G	A	
T	N	H	B	Acetate
U	O	I	C	Adipate
Z	P	K	E	Ag
R	L	F	Am	Al

Include Database

Include	Database
☒	thermochimie11a.dat

Selected Primary Entities/ Master Species

Incl.	Primary entity	Phase	Input variable	Fix log activity	Log activity	Concentration	Phase	Expression
☒	CO2[g]	gas		☒	-3.39			
☒	Ca	tot		☐		0.1	tot	
☒	Cl	tot		☐		0.19	tot	
☒	E	tot	pe	☒	-7.5			E.logact = -pe
☒	H	tot	pH	☒	-8.0			H.logact = -pH
☒	O	tot	H2O.logact	☒	0.0			O.logact = H2O.logact

☐ Balance Charge pe

No input file or directory selected.

chemistry1.inp

Read Write Text View

```
***** Version: 21 August 2024 07:58 *****
//*****
//NxDigits: 12 // number of decimal places in reaction coefficients.
//@rewriteReactions: //indicates that reactions are rewritten in terms of primary entities.
//@include_molar_volume_parameters: //include molar volume parameters from phreeqc database in species definitions.

//***** The database file(s) *****
// This section lists the database files that are used by the interactive chemistry editor
// The database file is not used during the calculations!
// Syntax for local files: @database: minteq.txt
// Syntax for files on the internet: @database: www.meeussen.nl/orchestra/minteqv4.txt

@database: PHREEQC_SIT_e- ThermoChimie_v12a.dat
@database: adsorption.txt
//***** End of the database file(s) *****

//***** Extra text 0 *****

@class: extra_text_0() {
@scan: objects2024.txt

// just an example how to solve arbitrary equations in ORCHESTRA
// Define 2 variables

//@Var: x .2
//@Var: y 9

// Define equation to calculate y from x
//@Calc: (1, "y = (x^2+4)")
```

No input file or directory selected.

ORCHESTRA: Selection of reactions from database

ORCHESTRA-Composer (Running on Windows 10 with 22 Oracle Corporation, using 16 processing cores)

File Run Tools Help

chemistry1.inp

Read Write GUI View

Primary entities/ Master Species Phases & Reactions Variables Adsorption models Activity correction Settings Predominance Diagram Output selector

Phase Hierarchy and Entities

Formation reactions in phase: **diss**, depending on primary entity: **all** and: **all** Hide unselected ☐ Alphabetical ☒ **Select all listed** **Select all recursively** **Unselect all**

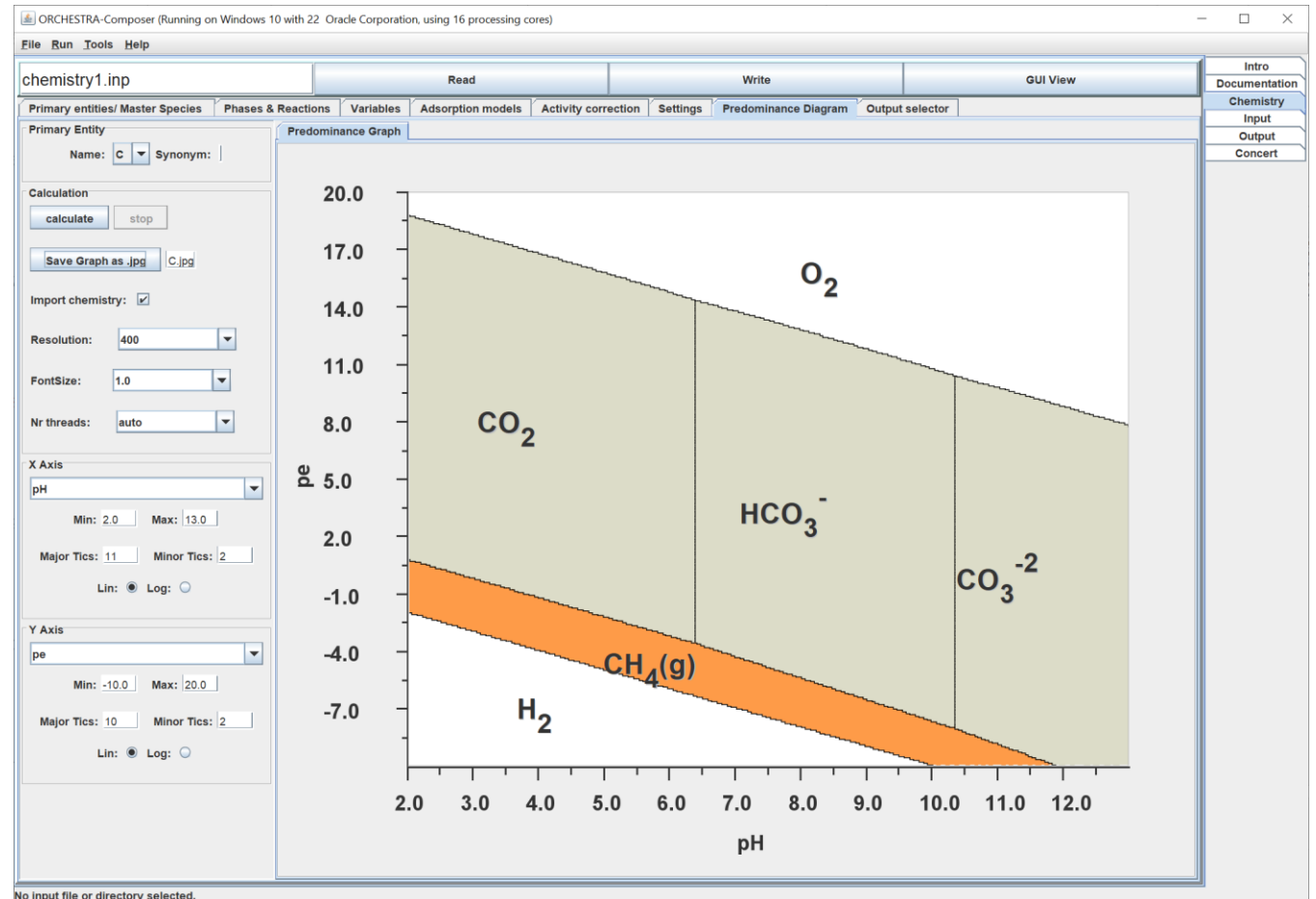
Incl.	Name	Log K (25C)	Phase	Coef.	Reactant	Coef.	Reactant	Coef.	Reactant
☐	CO	11.60000	diss	1.0	CO3-2	4.0	H+	-2.0	H2O
☒	CO2	-1.468000	diss	1.0	CO2[g]				
☒	CO3-2	-16.68100	diss	1.0	CO2	-2.0	H+	1.0	H2O
☒	Ca+2	0.0	diss	1.0	Ca				
☒	CaCO3	3.224000	diss	1.0	CO3-2	1.0	Ca+2		
☒	CaHCO3+	11.43500	diss	1.0	CO3-2	1.0	Ca+2	1.0	H+
☒	CaOH+	-12.78000	diss	1.0	Ca+2	-1.0	H+	1.0	H2O
☐	Ca[HCO3]+	11.43000	diss	1.0	CO3-2	1.0	Ca+2	1.0	H+
☐	Ca[OH]+	-12.78000	diss	1.0	Ca+2	-1.0	H+	1.0	H2O
☒	Cl-	0.0	diss	1.0	Cl				
☐	Cl2	-47.21000	diss	2.0	Cl-	-2.0	e-		
☐	ClO4-	-1.877900e2	diss	1.0	Cl-	-8.0	H+	4.0	H2O
☒	H+	0.0	diss	1.0	H				
☒	H2	-3.150000	diss	2.0	E	2.0	H		
☒	HCO3-	10.32900	diss	1.0	CO3-2	1.0	H+		
☒	O2	-86.08000	diss	-4.0	E	-4.0	H	2.0	O
☒	OH-	-13.99475	diss	-1.0	H	1.0	O		
☒	[CO2]2	-1.736241	diss	2.0	CO2				
☒	e-	0.0	diss	1.0	E				

No input file or directory selected.

Intro Documentation Chemistry Input Output Concert

Stability/predominance diagrams

- 2 input variables (e.g. pH pe)
- Calculate equilibrium at each location
- Indicate predominant species/form (with highest concentration)



Defining a chemical system

- Selection of primary entities
- Only independent entities can be selected (automatically)
- H, O, and H₂O represent water (from PHREEQC database)
- Additional components primary entities can be selected from database
- Each primary entity -> fixed log activity or fixed total mass balance
- Species can be selected
- Reactions are defined hierarchically but can be rewritten in terms of primary entities
- Activity correction
- Charge balance
- Predominance diagram

Format input files

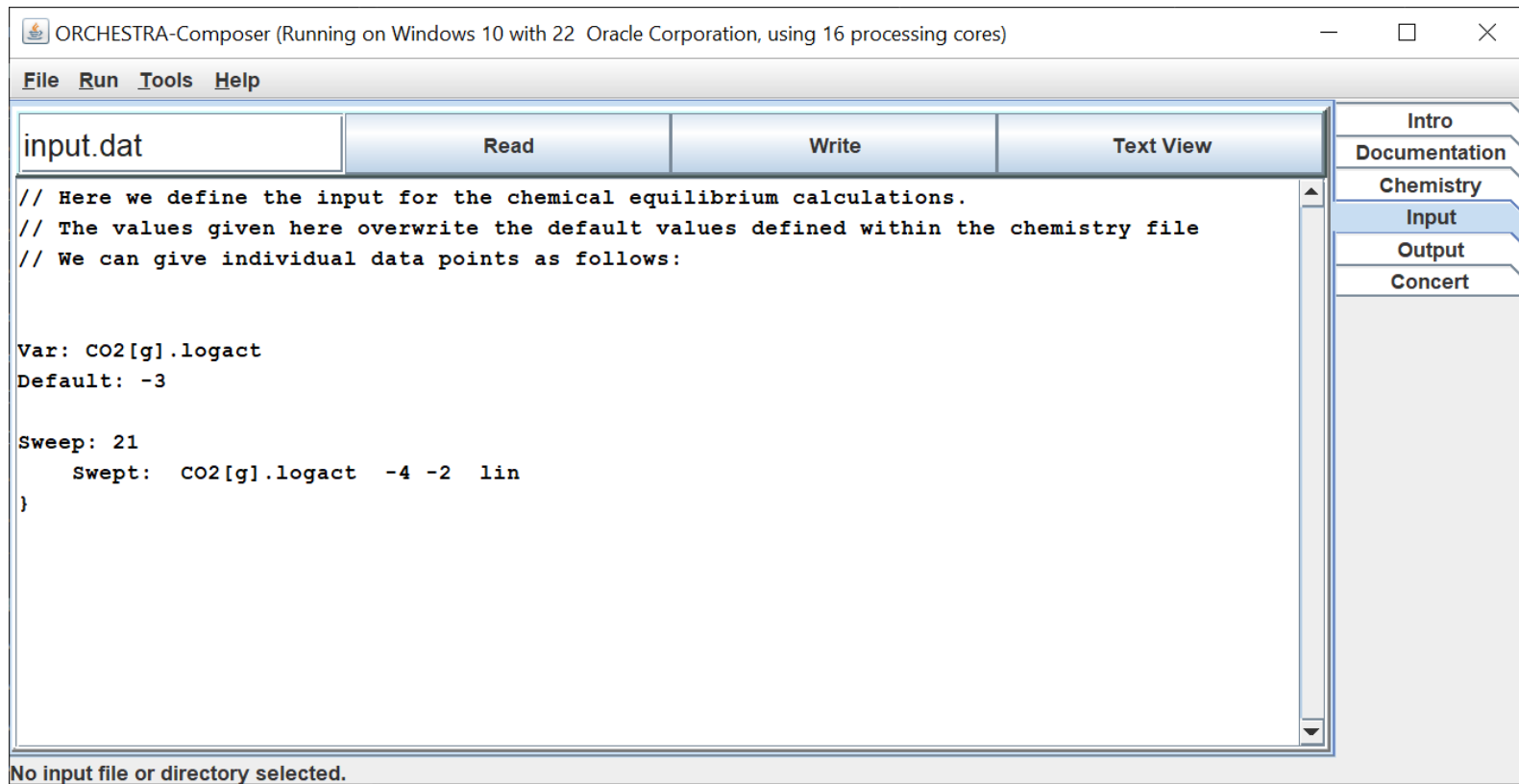
- User defined list of variables (column headers)
- User defined set of datapoints (rows)
- Can be pasted directly from Excel
- Order columns not important
- Values given in input overwrite default values in chemistry file

Var:	pH	pe	Ca.tot	.	.	.	.
Data:	7	8.3	1e-3	.	.	.	.
Data:	4	5.7	1.4e-2	.	.	.	.
.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.

Serial calculations using a loop / sweep

```
Sweep{ 21
  Swept: pH 2 12 lin
  Swept: pe 16 6 lin
}
```


ORCHESTRA: Input text file



Format output files

Basic output

Var: pH pe Ca.tot

- Similar to format input files
- **Output variables are defined in output file itself!!!!**
- First row column headers of output file defines output variables and their order
- Output can be directly pasted into Excel to create graphs

This is very flexible, (any variable can be requested for output) **but** you need to know the names of the variables you want!

(consequence of modular orchestra structure: output module has no knowledge of chemistry)

Other options

- PHREEQC type output (see demo)
- GUI can help with selection of variables

ORCHESTRA: output

ORCHESTRA-Composer (Running on Windows 10 with 22 Oracle Corporation, using 16 processing cores)

File Run Tools Help

output.dat

Read

Write

Text View

@PHREEQCOutput: chemistry1.inp

//Var: CO2[g].logact pH chargebalance

}

@format()

-----Description of solution-----

Cellnumber = 0.0

pH = 8.00000000

pe = 7.50000000

Activity of water = 9.96529e-1

Ionic strength = 1.23024263e-1

Total number of iterations = 33.0000000

-----Distribution of aqueous species-----

Species	concentration (Mol/l)		Log Activity		Activity	log
CO2[g]	2.18545926e-4					
CO2	3.40408190e-6	-5.46800000	3.40408e-6		0.0	
CO3-2	1.95255042e-6	-6.15051007	7.07115e-7		-4.41112e-1	
CaCO3	5.54625713e-6	-5.25600000	5.54626e-6		0.0	
CaHCO3+	1.17115481e-5	-5.04500000	9.01571e-6		-1.13614e-1	
HCO3-	1.95931488e-4	-3.82151007	1.50831e-4		-1.13614e-1	
[CO2]2	2.12695846e-13	-12.6722410	2.12696e-13		0.0	
Ca	1.39752603e-2					
Ca+2	1.39579019e-2	-2.32948993	4.68285e-3		-4.74310e-1	
CaCO3	5.54625713e-6	-5.25600000	5.54626e-6		0.0	
CaHCO3+	1.17115481e-5	-5.04500000	9.01571e-6		-1.13614e-1	
CaOH+	1.00603782e-7	-7.11100000	7.74462e-8		-1.13614e-1	
Cl	1.90000000e-1					
Cl-	1.90000000e-1	-8.50573580e-1	1.41067e-1		-1.29327e-1	
E	-3.30399859e-24					
H2	7.07945784e-35	-34.1500000	7.07946e-35		0.0	
O2	8.25999647e-25	-24.0830201	8.26000e-25		0.0	
e-	0.0	-7.50000000	3.16228e-8		-1.11111e-111	
H	-2.24081772e-4					
CO3-2	1.95255042e-6	-6.15051007	7.07115e-7		-4.41112e-1	
CaCO3	5.54625713e-6	-5.25600000	5.54626e-6		0.0	
CaHCO3+	1.17115481e-5	-5.04500000	9.01571e-6		-1.13614e-1	
CaOH+	1.00603782e-7	-7.11100000	7.74462e-8		-1.13614e-1	
H+	1.22411522e-8	-8.00000000	1.00000e-8		-8.78223e-2	
H2	7.07945784e-35	-34.1500000	7.07946e-35		0.0	
HCO3-	1.95931488e-4	-3.82151007	1.50831e-4		-1.13614e-1	
O2	8.25999647e-25	-24.0830201	8.26000e-25		0.0	

No input file or directory selected.

ORCHESTRA-Composer (Running on Windows 10 with 22 Oracle Corporation, using 16 processing cores)

File Run Tools Help

output.dat

Read

Write

Text View

```
@class: format(){
  //Output_at: (time, from:, 0, to:, 10, steps:, 100 )
  //@PHREEQCOutput: chemistry1.inp
```

```
Var: CO2[g].logact pH chargebalance
```

```
}
```

```
@format()
```

```
// The column headers in this file can be edited and determine the output for the next run.
```

Var:	CO2[g].logact	pH	chargebalance
Data:	-4.00000000	8.00000000	-1.62273561e-1
Data:	-3.90000000	8.00000000	-1.68489838e-1
Data:	-3.80000000	8.00000000	-1.73289798e-1
Data:	-3.70000000	8.00000000	-1.77032545e-1
Data:	-3.60000000	8.00000000	-1.79981051e-1
Data:	-3.50000000	8.00000000	-1.82331701e-1
Data:	-3.40000000	8.00000000	-1.84234121e-1
Data:	-3.30000000	8.00000000	-1.85804861e-1
Data:	-3.20000000	8.00000000	-1.87137100e-1
Data:	-3.10000000	8.00000000	-1.88307772e-1
Data:	-3.00000000	8.00000000	-1.89383006e-1
Data:	-2.90000000	8.00000000	-1.90422493e-1
Data:	-2.80000000	8.00000000	-1.91483204e-1
Data:	-2.70000000	8.00000000	-1.92622771e-1
Data:	-2.60000000	8.00000000	-1.93902812e-1
Data:	-2.50000000	8.00000000	-1.95392409e-1
Data:	-2.40000000	8.00000000	-1.97171979e-1
Data:	-2.30000000	8.00000000	-1.99337779e-1
Data:	-2.20000000	8.00000000	-2.02007330e-1
Data:	-2.10000000	8.00000000	-2.05326097e-1
Data:	-2.00000000	8.00000000	-2.09475855e-1

No input file or directory selected.

Are there any questions about:

- defining a chemical system or
- defining input output format files?

Part II ORCHESTRA chemical model structure

ORCHESTRA chemical models are composed from a small set of basic building blocks or object classes

This “**object oriented**” approach:

- Ensures **consistency** and compatibility between models
- Makes model definitions very **compact**

Have a look in the **objects2024.txt** file (the chemical model library)

Phases

Entities

Reactions

Equations

Part II ORCHESTRA chemical model structure

Basic model building blocks

Equations	To solve arbitrary unknown – equation combinations
Entities	An abstract “species” with .con and .logact
Phases	Hierarchical system compartments in which entities can be present
Reactions	Reactions between entities to create new entities

Phases

Entities

Reactions

Equations

ORCHESTRA model building blocks: Entities

- Entities are a kind of abstract chemical “species”
- Each entity exists in a certain “phase” (with its own mass balance units)
- Each entity has the variables: <name>.logact <name>.con
- **All entity** types are automatically **compatible** with “**reactions**” and can be used as reactants.

Examples of entity types:

Entity type	represents	Present in phase	.logact	.con
@species	aqueous species	diss	Log activity	Concentration (mol/l)
@mineral	Precipitating mineral	min	Saturation index	Concentration (mol/l)
@gas	gas	gas	Partial pressure	Partial pressure
@surface_site	Surface site	<surface>	Empty fraction	Empty fraction
@surface_species	Adsorbed species	<site>	fraction	fraction
@e_layer	Electrostatic plane	<surface>	Boltzmann factor	0
@entity(e-, diss, 0)	electron	diss	-pe	0

ORCHESTRA Phases

- Phases represent system compartments with local mass balances for **primary entities** (e.g. tot, diss, liter, ads)
- Phases are connected **hierarchically** via **conversion** factors
- Parent phases contain sum of mass of all child phases

Example:

Ca.tot will contain total Ca in system in all forms **Ca.liter** x **watervolume**

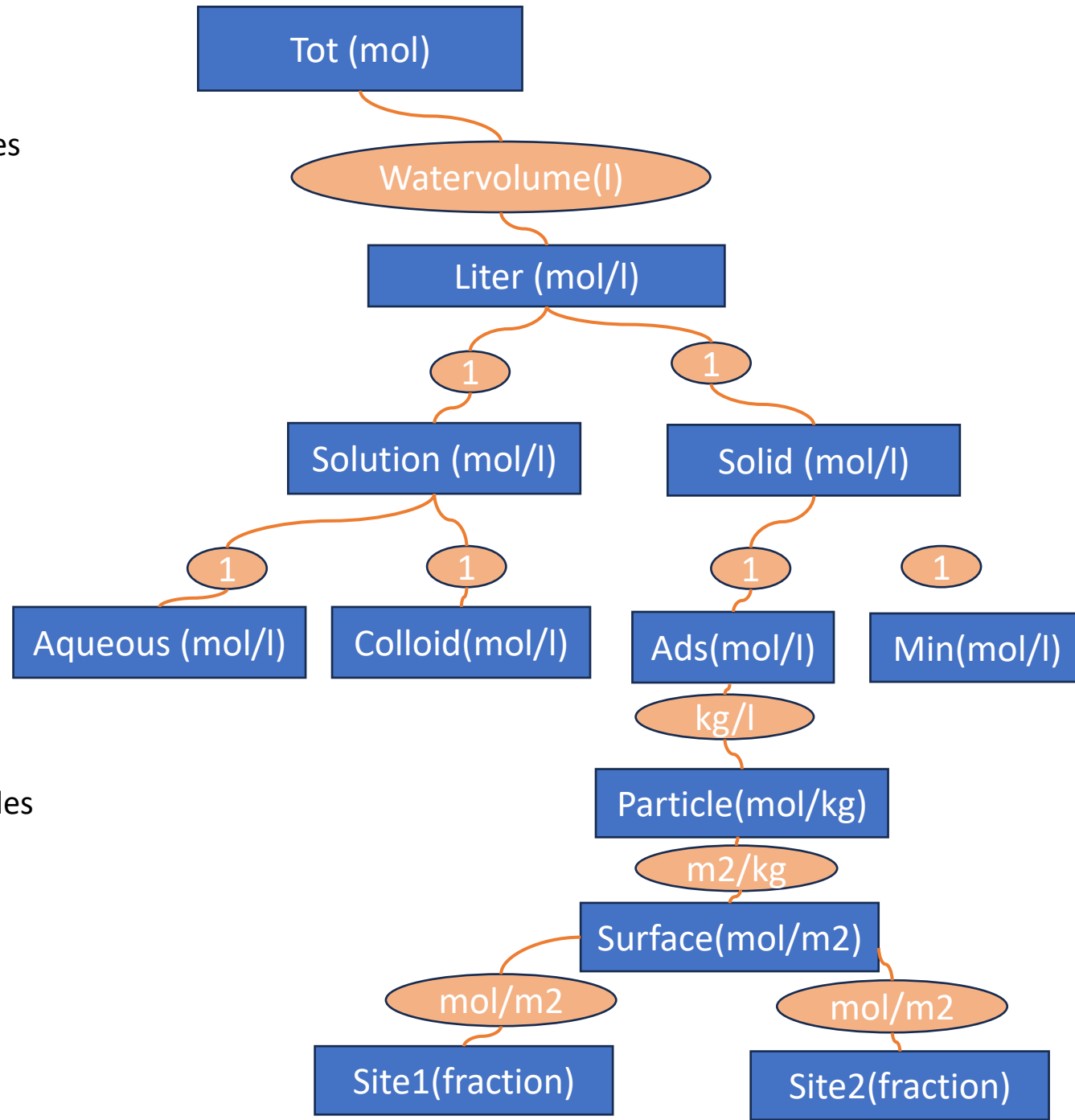
Ca.liter will contain total Ca in mol/l

Ca.solid will contain sum of all solid Ca in mol/l

Ca.ads will contain sum of all adsorbed Ca mol/l

Ca.Site1 will contain all adsorbed Ca to site 1 (fraction)

The phases enable hierarchical mass balances are useful for output, but also for communication with external codes and for creating adsorption models.



ORCHESTRA Phases

- Phases represent system compartments with local mass balances for **primary entities** (e.g. tot, diss, liter, ads)
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Example:

Ca.tot will contain total Ca in system in all forms **Ca.liter** x **watervolume**

Ca.liter will contain total Ca in mol/l

Ca.solid will contain sum of all solid Ca in mol/l

Ca.ads will contain sum of all adsorbed Ca mol/l

Ca.Site1 will contain all adsorbed Ca to site 1 (fraction)

The phases enable hierarchical mass balances are useful for output, but also for communication with external codes and for creating adsorption models.

The screenshot displays the ORCHESTRA-Composer software interface. The title bar indicates it is running on Windows 10 with version 20.0.2, using 8 processing cores. The main window features a menu bar (File, Run, Tools, Help) and a toolbar with buttons for 'Read', 'Write', and 'GUI View'. Below the toolbar, there are tabs for 'Activity correction', 'Settings', 'Predominance Diagram', 'Output selector', 'Primary entities: Master Species', 'Phases & Reactions', 'Variables', and 'Adsorption models'. The 'Phases & Reactions' tab is active, showing a 'Phase Hierarchy and Entities' tree. The tree structure is as follows:

- Reactions in all phases
 - tot (1)
 - liter (watervolume)
 - solution (1)
 - diss (1)
 - colloid (1)
 - DHA_part (1)
 - HA_part (DHA_kgl)
 - DFA_part (1)
 - FA_part (DFA_kgl)
 - solid (1)
 - min (1)
 - ads (1)
 - CLAY_part_I (1)
 - CLAY_part (CLAY_kgkg*SL)
 - HFO_part_I (1)
 - HFO_part (HFO_kgkg*SL)
 - Goethite_part_I (1)
 - Goethite_part (Goethite_kgkg*SL)
 - SHA_part_I (1)
 - HA_part (SHA_kgkg*SL)
 - SFA_part_I (1)
 - FA_part (SFA_kgkg*SL)
 - exchange (1)
- gas_I (1)
 - gas ((gasvolume/22.4))

The right sidebar contains a navigation pane with links to 'Intro', 'Documentation', 'Chemistry', 'Input', 'Output', 'Output2', 'Output3', and 'Concert'. The 'Chemistry' link is selected. At the bottom of the window, a status bar reads 'No input file or directory selected.'

ORCHESTRA Phases

- Phases are user definable

```
@phase(tot)
@phase(kg)
@phase(liter)
@link_phase(liter, tot, "watervolume")
@link_phase(liter, kg, "1/SL")
@phase(solution)
@link_phase(solution, liter, "1")
@phase(diss)
@link_phase(diss, solution, "1")
@phase(colloid)
@link_phase(colloid, solution, "1")
@phase(DHA_part)
@link_phase(DHA_part, colloid, "1")
@phase(DFA_part)
@link_phase(DFA_part, colloid, "1")
@phase(solid)
@link_phase(solid, liter, "1")
```

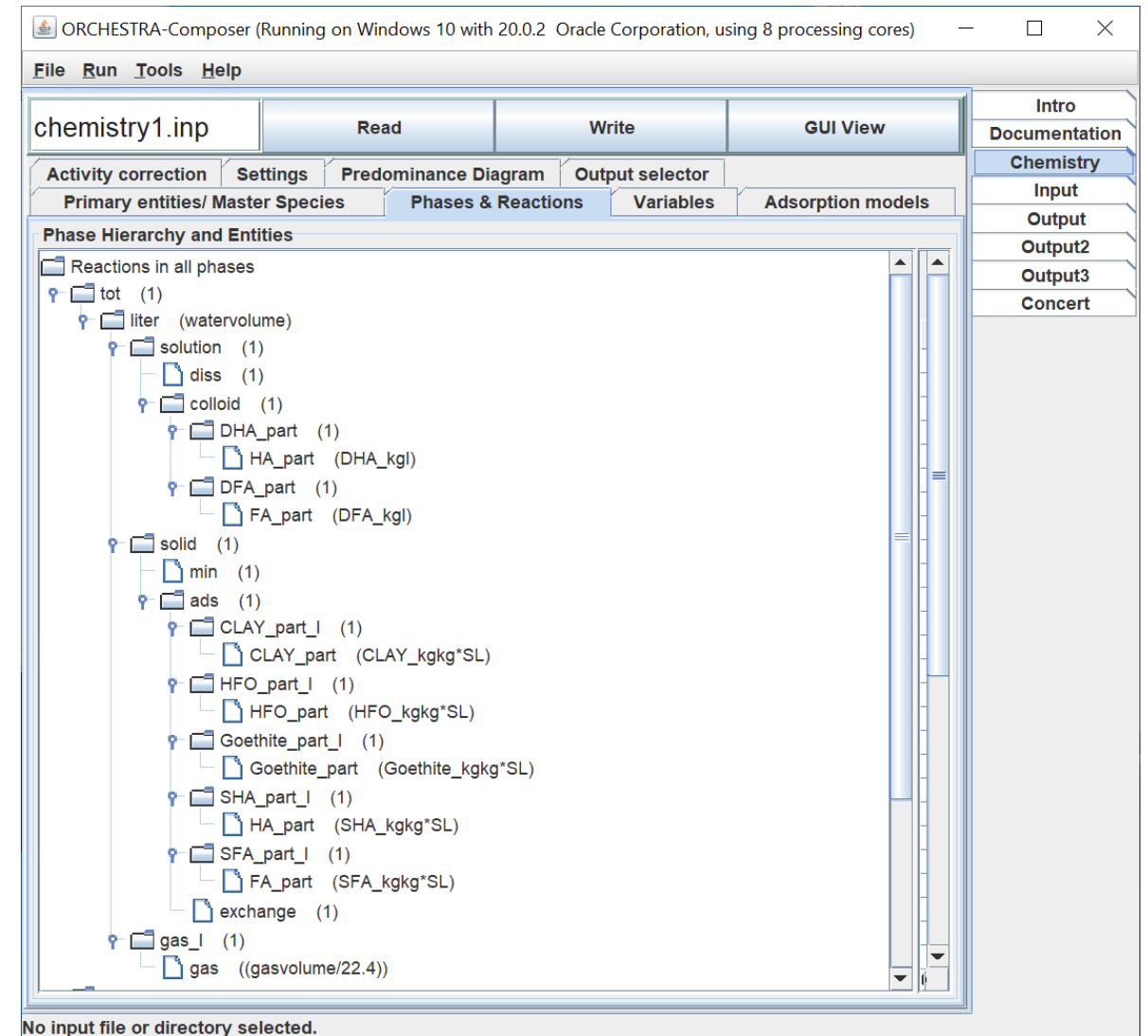
The screenshot displays the ORCHESTRA-Composer software interface. The title bar indicates it is running on Windows 10 with version 20.0.2, using 8 processing cores. The main window features a menu bar (File, Run, Tools, Help) and a toolbar with buttons for 'chemistry1.inp', 'Read', 'Write', and 'GUI View'. Below the toolbar is a tabbed interface with tabs for 'Activity correction', 'Settings', 'Predominance Diagram', 'Output selector', 'Primary entities/ Master Species', 'Phases & Reactions' (which is currently selected), 'Variables', and 'Adsorption models'. On the right side, there is a vertical sidebar with buttons for 'Intro', 'Documentation', 'Chemistry' (highlighted), 'Input', 'Output', 'Output2', 'Output3', and 'Concert'. The main workspace shows a 'Phase Hierarchy and Entities' tree. The tree structure is as follows: 'Reactions in all phases' (root) -> 'tot (1)' -> 'liter (watervolume)' -> 'solution (1)' -> 'diss (1)' -> 'colloid (1)' -> 'DHA_part (1)' (with sub-entity 'HA_part (DHA_kgl)') -> 'DFA_part (1)' (with sub-entity 'FA_part (DFA_kgl)') -> 'solid (1)' -> 'min (1)' -> 'ads (1)' -> 'CLAY_part_I (1)' (with sub-entity 'CLAY_part (CLAY_kgkg*SL)') -> 'HFO_part_I (1)' (with sub-entity 'HFO_part (HFO_kgkg*SL)') -> 'Goethite_part_I (1)' (with sub-entity 'Goethite_part (Goethite_kgkg*SL)') -> 'SHA_part_I (1)' (with sub-entity 'HA_part (SHA_kgkg*SL)') -> 'SFA_part_I (1)' (with sub-entity 'FA_part (SFA_kgkg*SL)') -> 'exchange (1)' -> 'gas_I (1)' -> 'gas ((gasvolume/22.4))'. The status bar at the bottom of the window displays the message 'No input file or directory selected.'

ORCHESTRA Phases

The phases enable hierarchical mass balances

Useful for

- Input/Output
- For communication with external (transport) codes
- For creating adsorption models



ORCHESTRA Primary Entities

- **Primary entities** are similar to **master species** (PHREEQC) or **components** (ECOSAT)
- Primary entities can be **elements** e.g. Al (PHREEQC) but also **aqueous ions** e.g. Al⁺³ (ECOSAT)
- In **PHREEQC/ECOSAT** primary entities are **defined in thermodynamic database**
- In ORCHESTRA **any** independent **entity** can be made **primary** (The GUI will check independency)

ORCHESTRA model building blocks: Primary Entities

- Primary entities have a .logact and .con variable, but they **additionally** have a **mass balance variable** for **each** system phase
- For example when Al is selected as primary entity:
 - Al.diss contains the **sum** of al Al forms in the **diss** phase (e.g. sum of Al+3, Al[OH]2+ Al[OH]+ etc.)
 - Al.tot contains the **sum** of al Al forms in the **tot** phase
- For example when Al+3 is selected as primary entity:
 - Al+3.diss contains the **sum** of al Al+3 forms in the **diss** phase (e.g. sum of Al+3, Al[OH]2+ Al[OH]+ etc.)
 - Al+3.tot contains the **sum** of al Al+3 forms in the **tot** phase

To calculate the system for each primary entity we need as input a **given .logact**, or **given cumulative mass balance** in a certain system phase.

When a primary entity is selected, the GUI shows all possible dependent entities that can be formed and selected from the set of primary entities.

ORCHESTRA demo

- Install Java on your computer
- Download complete zipped folder from:

www.meeussen.nl/orchestra/getstarted.zip

Adsorption example:

www.meeussen.nl/orchestra/getstarted2.zip

ORCHESTRA adsorption models, how to use them

- Include the file “adsorption_2022.txt” as database in the chemistry file

ORCHESTRA-Composer (Running on Windows 10 with 20.0.2 Oracle Corporation, using 8 processing cores)

File Run Tools Help

chemistry1.inp Read Write GUI View

Adorption models Activity correction Settings Predominance Diagram Output selector

Primary entities/ Master Species Phases & Reactions Variables Extra Text

Selectable Primary Entities/ Master Species in Database

tot diss min gas

M A Acetate
N B Adipate
O C Ag
P E Al
R F Alkalinity
S G Am
T H Am[+2]
U I Am[+3]
Z K Am[+4]
L Am[+5]

| Include | Database |
|-------------------------------------|-------------------------------------|
| ☒ | PHREEQC_SIT_e_ThermoChimie_v12a.dat |
| ☒ | adsmodels2022.txt |

Add database file

Selected Primary Entities/ Master Species

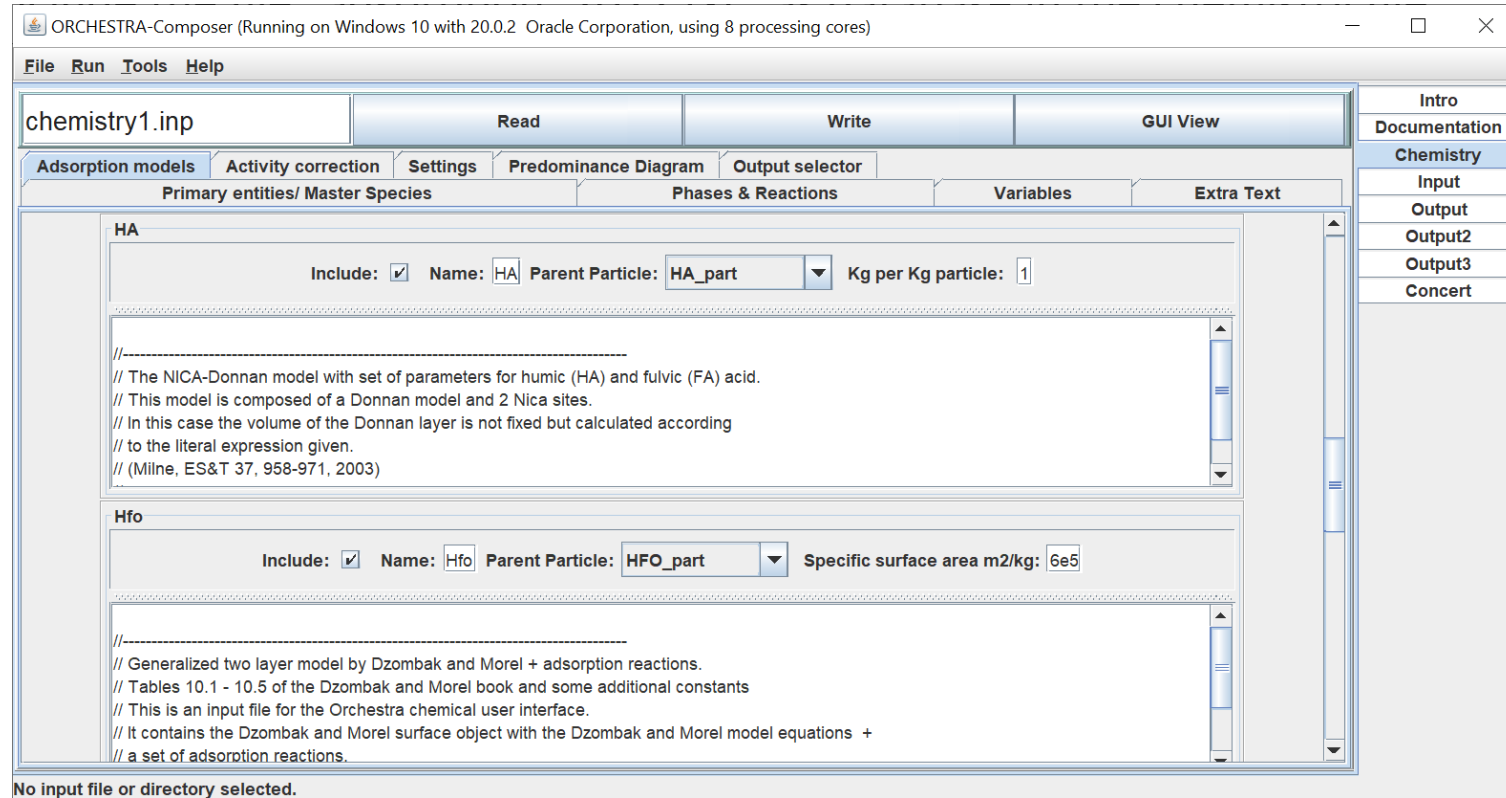
| Incl. | Primary entity | Phase | Input variable | Fix log activity | Log activity | Concentration | Phase | Expression |
|-------------------------------------|----------------|-------|----------------|-------------------------------------|--------------|---------------|-------|------------|
| ☒ | Ca | tot | | ☐ | | 1.0E-9 | tot | |
| ☒ | F | tot | | ☒ | 7.0 | | | |

☐ Balance Charge pe

No input file or directory selected.

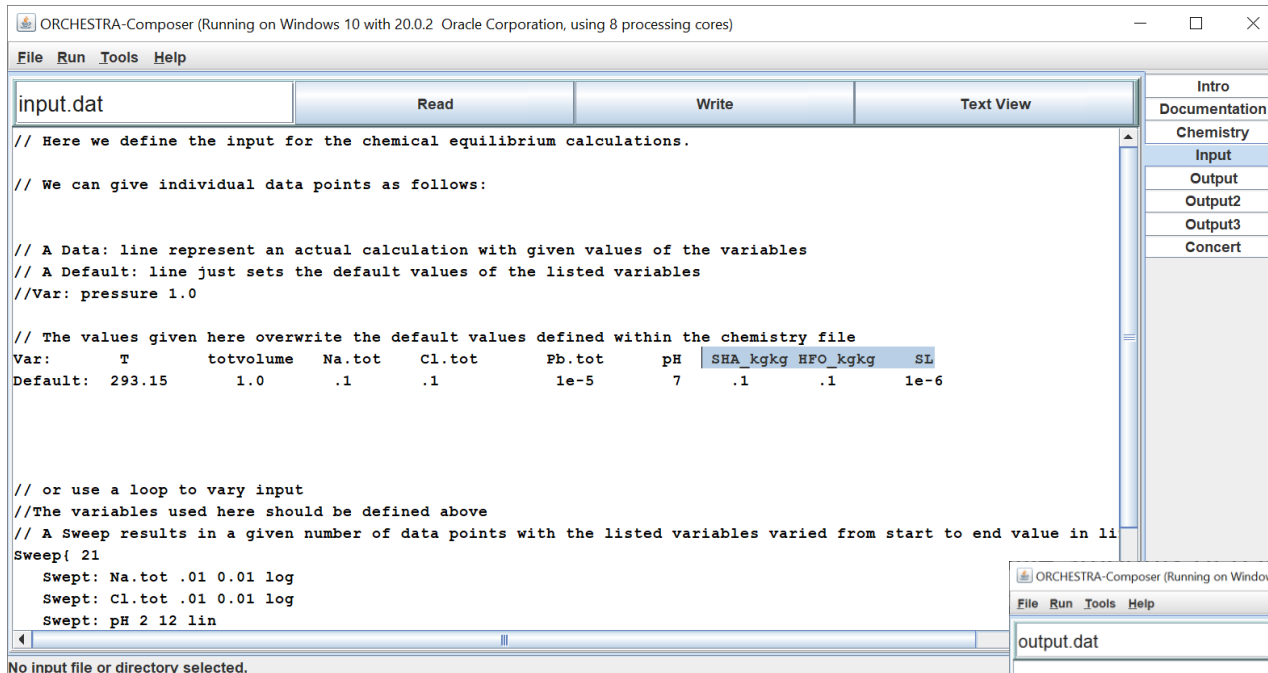
ORCHESTRA adsorption models, how to use them

- Select and connect to appropriate particle



ORCHESTRA adsorption models, how to use them

- Give amounts of particle in input file



ORCHESTRA-Composer (Running on Windows 10 with 20.0.2 Oracle Corporation, using 8 processing cores)

File Run Tools Help

input.dat Read Write Text View

```
// Here we define the input for the chemical equilibrium calculations.

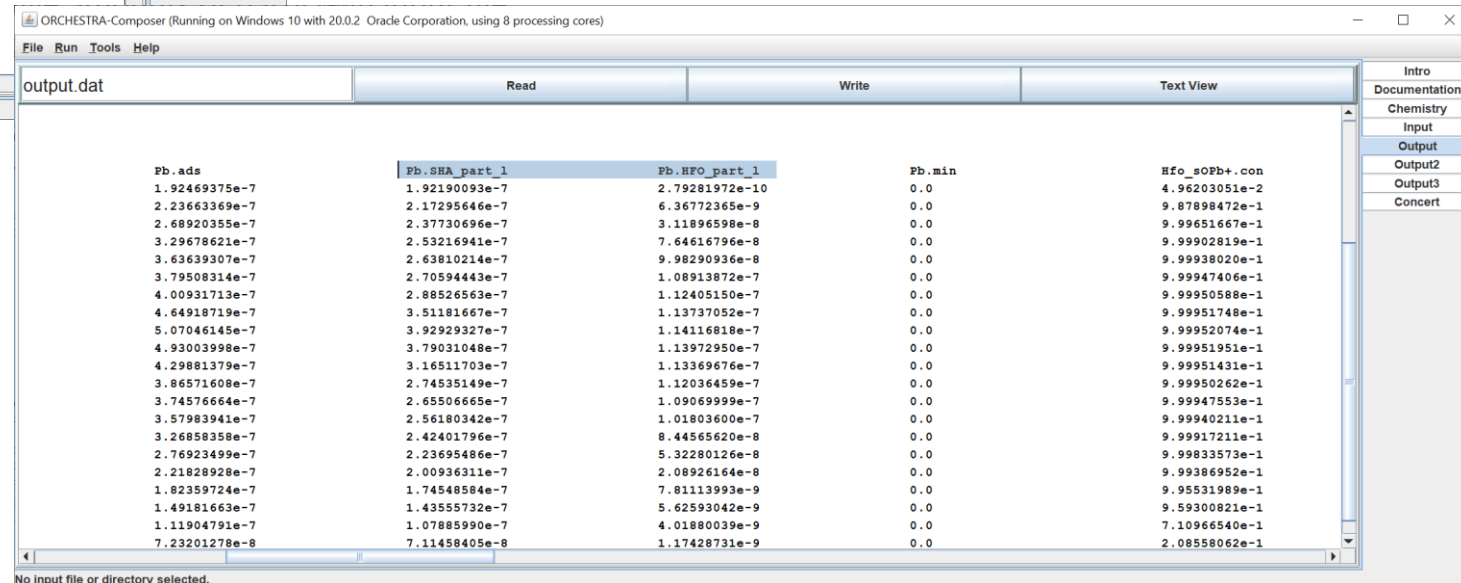
// We can give individual data points as follows:

// A Data: line represent an actual calculation with given values of the variables
// A Default: line just sets the default values of the listed variables
//Var: pressure 1.0

// The values given here overwrite the default values defined within the chemistry file
Var:      T      totvolume  Na.tot  Cl.tot  Pb.tot  pH  SHA_kgkg  HFO_kgkg  SL
Default: 293.15      1.0      .1      .1      1e-5  7    .1      .1      1e-6

// or use a loop to vary input
//The variables used here should be defined above
// A Sweep results in a given number of data points with the listed variables varied from start to end value in li
Sweep{ 21
  Swept: Na.tot .01 0.01 log
  Swept: Cl.tot .01 0.01 log
  Swept: pH 2 12 lin
```

No input file or directory selected.



ORCHESTRA-Composer (Running on Windows 10 with 20.0.2 Oracle Corporation, using 8 processing cores)

File Run Tools Help

output.dat Read Write Text View

| Pb.ads | Pb.SHA_part_1 | Pb.HFO_part_1 | Pb.min | Hfo_sOPb+.con |
|---------------|---------------|----------------|--------|---------------|
| 1.92469375e-7 | 1.92190093e-7 | 2.79281972e-10 | 0.0 | 4.96203051e-2 |
| 2.23663369e-7 | 2.17295646e-7 | 6.36772365e-9 | 0.0 | 9.87898472e-1 |
| 2.68920355e-7 | 2.37730696e-7 | 3.11896598e-8 | 0.0 | 9.99651667e-1 |
| 3.29678621e-7 | 2.53216941e-7 | 7.64616796e-8 | 0.0 | 9.99902819e-1 |
| 3.63639307e-7 | 2.63810214e-7 | 9.98290936e-8 | 0.0 | 9.99938020e-1 |
| 3.79508314e-7 | 2.70594443e-7 | 1.08913872e-7 | 0.0 | 9.99947406e-1 |
| 4.00931713e-7 | 2.88526563e-7 | 1.12405150e-7 | 0.0 | 9.99950588e-1 |
| 4.64918719e-7 | 3.51181667e-7 | 1.13737052e-7 | 0.0 | 9.99951748e-1 |
| 5.07046145e-7 | 3.92929327e-7 | 1.14116818e-7 | 0.0 | 9.99952074e-1 |
| 4.93003998e-7 | 3.79031048e-7 | 1.13972950e-7 | 0.0 | 9.99951951e-1 |
| 4.29881379e-7 | 3.16511703e-7 | 1.13369676e-7 | 0.0 | 9.99951431e-1 |
| 3.86571608e-7 | 2.74535149e-7 | 1.12036459e-7 | 0.0 | 9.99950262e-1 |
| 3.74576664e-7 | 2.65506665e-7 | 1.09069999e-7 | 0.0 | 9.99947553e-1 |
| 3.57983941e-7 | 2.56180342e-7 | 1.01803600e-7 | 0.0 | 9.99940211e-1 |
| 3.26858358e-7 | 2.42401796e-7 | 8.44565620e-8 | 0.0 | 9.99917211e-1 |
| 2.76923499e-7 | 2.23695486e-7 | 5.3280126e-8 | 0.0 | 9.99833573e-1 |
| 2.21828928e-7 | 2.00936311e-7 | 2.08926164e-8 | 0.0 | 9.99386952e-1 |
| 1.82359724e-7 | 1.74548584e-7 | 7.81113993e-9 | 0.0 | 9.95531989e-1 |
| 1.49181663e-7 | 1.43555732e-7 | 5.62593042e-9 | 0.0 | 9.59300821e-1 |
| 1.11904791e-7 | 1.07885990e-7 | 4.01880039e-9 | 0.0 | 7.10966540e-1 |
| 7.23201278e-8 | 7.11458405e-8 | 1.17428731e-9 | 0.0 | 2.08558062e-1 |

No input file or directory selected.

Are there any questions/comments?

Recap of today:

- ORCHESTRA, what's different (compared to ECOSAT, PHREEQC etc.)
- Download “install” ORCHESTRA
- Define chemical equilibrium system,
- Define text input, output
- How to work with “phases”, “entities”, “reactions”
- How to work with adsorption models

Thank you for your participation!

- Presentation will be sent to participants
- Next course Wednesday October 9 15:30-17:00
- Possible topics:
 - More on standard chemistry (gases, solid solutions, activity correction)
 - Adsorption models
 - Kinetics
 - Transport
 - Coupling with Python
 - Other based on you feed-back
- Please send feed-back to meeussen@nrg.eu